ELEC5622 · PROJECT 1

Classifying AD Patients and Normal Controls from Brain Images

Regional Grey Matter Volumes and Support Vector Machines

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Abstract

Regional grey matter volumes were used to classify Alzheimer's disease (AD) and normal control (NC) subjects from 50 T1-weighted MRI scans. Brain extraction, tissue segmentation and template-to-native registration produced grey matter volume features for the first 90 AAL regions. A StandardScaler and RBF support vector machine were evaluated using stratified five-fold cross-validation on 40 training subjects and then fitted to the complete training set. Mean cross-validation accuracy was 97.5%. On the 10 held-out subjects, accuracy was 90%, AD sensitivity was 83.3% and NC specificity was 100%. A post hoc training-only analysis identified differences in overall GM volume and class-associated image geometry. These observations, the small test set and residual preprocessing limitations restrict disease-specific interpretation and motivate independent validation with acquisition controls.

Keywords: Alzheimer's disease; structural MRI; grey matter volume; support vector machine.

Code and data availability: Preprocessing scripts, the SVM notebook, ROI volume tables and verification records.
https://github.com/Babe153/Project01_5622

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1. Aims and Background

1.1 Background and Rationale

Previous MRI studies have reported grey matter atrophy in multiple brain regions in patients with Alzheimer's disease (AD) [1]. These findings provide evidence of an association between brain structure and AD. Other research has used structural MRI and support vector machines (SVMs) to distinguish patients with AD from normal controls (NC) [2]. Based on this background, this project uses regional grey matter volumes as numerical features to investigate their usefulness for classifying AD and NC subjects.

A predefined AAL parcellation [3] reduces each three-dimensional scan to 90 regional measurements with consistent anatomical identities. This representation is compact enough to inspect and link to individual brain regions, although pooling within regions can discard local spatial detail. The processing stages determine what the features mean: non-brain removal limits unwanted tissue, segmentation identifies GM, and registration locates the same atlas regions in each subject. Errors at any stage can alter the classifier inputs. A 2024 review documents heterogeneous MRI-based classification tasks and study populations [4], so we assess this specified pipeline on the supplied split without assuming that published performance transfers to this dataset.

1.2 Project Aims

We aim to classify AD and NC subjects from T1-weighted MRI using 90 regional GM volumes and an SVM. Preprocessing constructs comparable anatomical measurements; training-only standardization and validation then assess their predictive use before testing ten held-out subjects. Lab 1 contributes the spatial operations and physical-volume interpretation. Lab 2 contributes supervised fitting, prediction and evaluation. Combining them makes the links between an image-processing choice, a numerical feature and a classification result explicit.

2. Contributions of Team Members

Table 1. Contributions of team members

Team member	Contributions	Share
Xinsheng Liu	Primarily responsible for code implementation and computational experiments, including completing the preprocessing scripts, adjusting BET parameters, extracting regional grey matter volumes, and implementing SVM training and evaluation. Also organized the code, execution records and output files.	50%
Yue Xia	Primarily responsible for reviewing the processing workflow and representative results, selecting and explaining figures, reviewing relevant literature, and writing and organizing the report. Also checked the consistency of the reported methods and results with the project materials.	50%

Both team members contributed equally to the project, with different primary responsibilities.

3. Methodology

3.1 Dataset and Experimental Design

We use the 50 T1-weighted MRI scans and AD/NC labels supplied by the instructor. The fixed training set, Data_00–Data_39, contains 21 NC and 19 AD subjects; the test set, Data_40–Data_49, contains 4 NC and 6 AD subjects. Each scan yields 90 GM volume features. Five-fold stratified cross-validation uses only the 40 training subjects, followed by final fitting on those subjects. The 10 test subjects are predicted with the fitted pipeline; their labels are used only for final evaluation, not fitting or parameter selection. The label table contains no demographic covariates, and we did not estimate intracranial volume for covariate adjustment.

The recorded preprocessing environment is an Ubuntu Linux VM with FSL 6.0.7.1 and NiftyReg 1.5.58. The MNI152 T1 brain template and AAL atlas are the supplied Packages files. Local classification uses Python 3.12, NumPy 2.5.3, pandas 3.0.6 and scikit-learn 1.9.1; the latter library versions are saved in the training record. The feature tables, BET settings and verification records accompany the implementation in the repository.

3.2 Overview of the Processing Pipeline

Figure 1 summarises the workflow from MRI preprocessing to regional grey matter volume extraction and SVM classification.

The workflow in Figure 1 begins with brain extraction and tissue segmentation of the original MRI scans to obtain grey matter masks and grey matter intensity images. Affine and nonlinear registration determine the transformations used to map the AAL atlas into each subject's native space. Each ROI mask is intersected with the grey matter mask, and grey matter volume is measured for the first 90 AAL regions. The resulting 90 volumes form one feature vector per subject. Feature standardization and SVM fitting are performed within each training fold for cross-validation, followed by fitting on all training subjects and prediction of the test set.

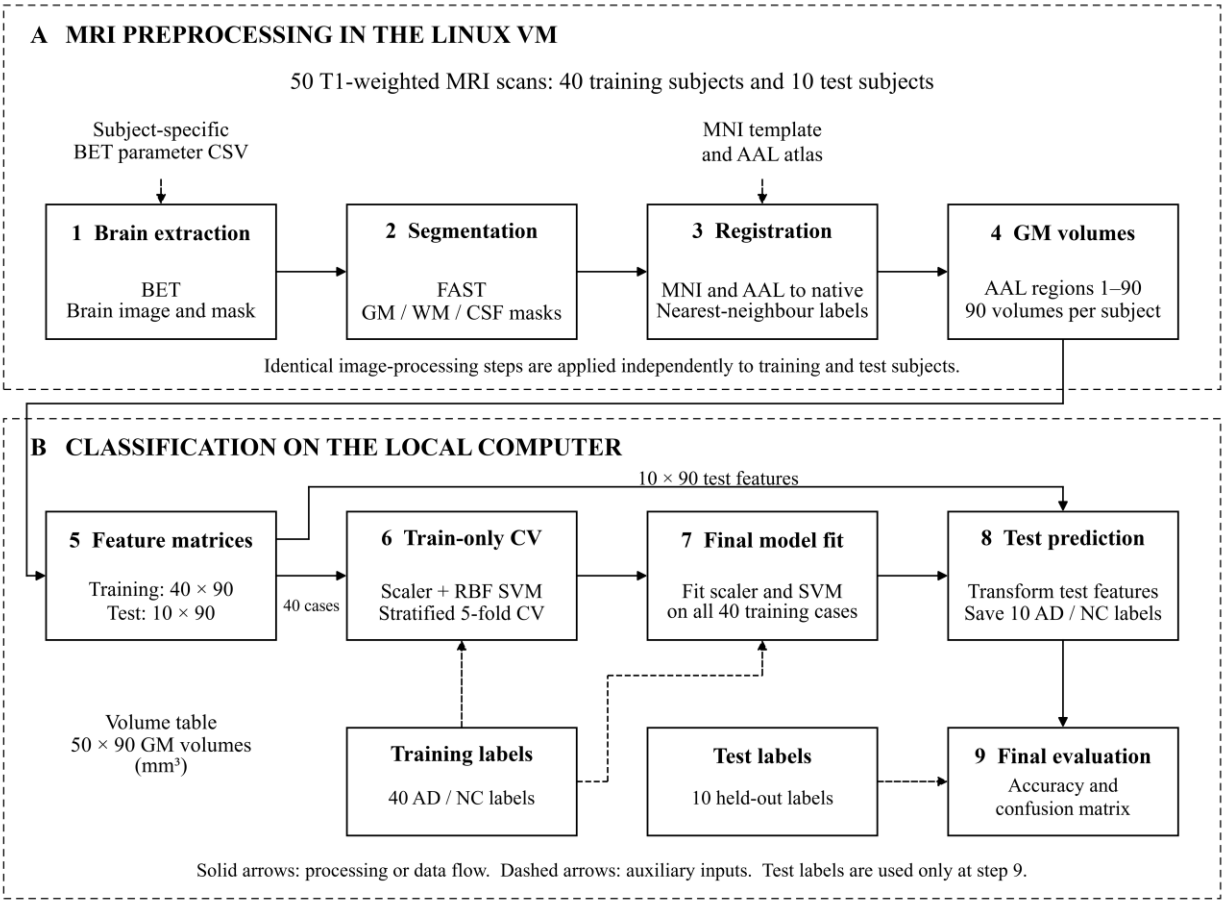


Figure 1. Processing and classification workflow. Dashed boundaries distinguish the Linux VM and local computer. Test features bypass model fitting; held-out test labels are used only after predictions have been saved.

3.3 Brain Extraction

The original T1-weighted MRI images contain both brain tissue and non-brain tissue. In this project, we use FSL's Brain Extraction Tool (BET) to remove non-brain tissue and reduce its influence on subsequent tissue segmentation. The script reads the settings for each subject from the parameter table and processes the images in batches. This step produces brain-extracted images and binary brain masks. The brain-extracted images are then used for tissue segmentation in the next step. BET uses a deformable surface to separate brain from non-brain tissue [5].

Perform brain extraction and generate a binary brain mask for Data_00:

```
bet Data/train/Data_00.nii Output/train/Data_00_brain.nii.gz \
  -f 0.3 -g 0 -c 109 165 82 -m
```

This command takes the original MRI of Data_00 as input and saves the brain-extracted image. The parameter ***-f*** sets the fractional intensity threshold to 0.3, while ***-g*** 0 applies no vertical threshold gradient. The parameter -c specifies the initial brain center at voxel coordinates (109, 165, 82). The option ***-m*** also produces a binary brain mask.

The command above uses the settings for Data_00. During batch processing, SkullStripping reads each subject's parameters from bet_parameters.csv and runs BET on the training and test images. The four main scripts run in the order SkullStripping, TissueSegmentation, Registration and Measurement. Measurement calls the supplied, unmodified CreateSeedMask helper to generate the regional masks. Appendix A lists representative commands for these processing stages.

BET settings were selected by inspecting sampled sagittal, coronal and axial boundaries for brain coverage and non-brain remnants. For Data_00, the adopted centre and threshold reduce the neck and facial remnants visible with the default settings while retaining the main brain outline in the inspected slices. Disease labels and classification scores were not the selection criteria. The parameter table records adopted settings rather than a global optimum; the retained Data_44 limitation is discussed in Section 5.2.

3.4 Tissue Segmentation and Grey Matter Image Generation

After brain extraction, FSL FAST segments the T1 images into grey matter, white matter and cerebrospinal fluid. Its hidden Markov random field represents dependence between neighbouring tissue labels, while expectation-maximization estimates the model from observed intensities [6]. Spatial context therefore contributes beyond an intensity histogram alone, but it does not supply manual anatomical truth. We use the instructor's three-class settings and export binary tissue masks. Multiplying the brain image by the GM mask produces a GM intensity image for inspection. Regional volume uses the binary GM mask, not the sum of MRI intensities or tissue probabilities. This hard assignment discards partial tissue membership at boundaries. Appendix A.1 gives the commands; Figure B2 shows the GM intensity image.

3.5 Image Registration and AAL Atlas Transformation

We register the supplied MNI brain template to each native brain with NiftyReg [7]: affine alignment handles global position, orientation and scale, followed by nonlinear refinement for local shape differences. The subject is the reference and the original template is the floating image in both stages (Appendix A.2). Applying the resulting transformation to AAL [3] with nearest-neighbour interpolation retains integer region identities. The GM mask stays on the native grid, where physical volume is counted without Jacobian modulation of a normalized GM image. This is regional native-volume analysis rather than voxel-based morphometry. Integer labels do not establish anatomical correspondence; alignment needs separate inspection.

3.6 Regional Grey Matter Volume Extraction

Inputs and ROI definition. Measurement uses the native-space AAL label image and binary GM mask produced by the preceding stages. Their voxel grids must match. The unchanged CreateSeedMask helper isolates each atlas label with thresholding and binarization; Measurement selects labels 1–90. Figure 2 illustrates label 1 for Data_00.

Restricting the GM mask. For each region, the binary ROI mask selects voxels from the binary GM mask. Only voxels that belong to both masks contribute to the measurement. This intersection excludes white matter and CSF from the regional count; image intensity is not summed.

Physical volume. We run fslstats on the GM mask, selecting the ROI with -k. Its -V output gives the non-zero voxel count followed by physical volume in mm³. Volume is the complete 3D intersection count multiplied by voxel volume from the image header. The displayed slice is illustrative; counting uses all slices. Appendix A.3 provides the command.

Feature export and checking. We save the second -V value and repeat for all 90 regions. Each CSV row contains one subject ID followed by ROI_1_mm3 through ROI_90_mm3, giving 40 × 90 training and 10 × 90 test matrices. For the illustrated case, 6109 GM voxels give 6523.223317 mm³, matching the stored ROI_1_mm3 value. IDs and disease labels are excluded from the feature matrix.

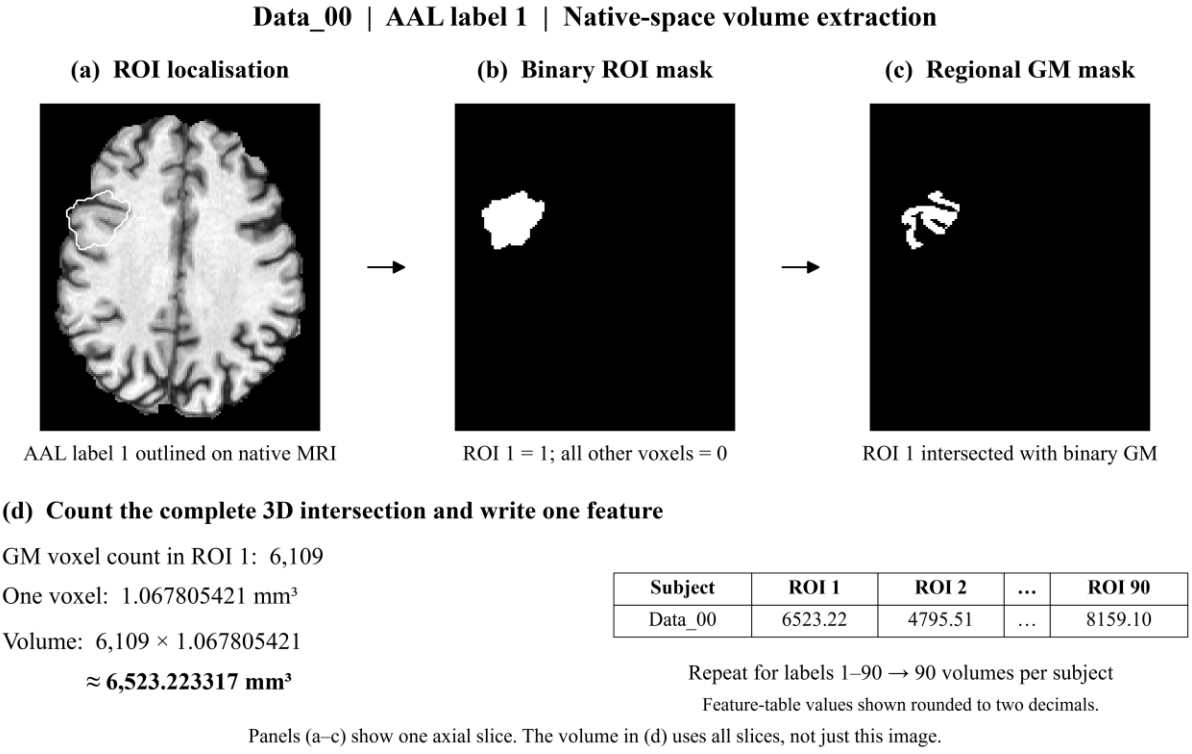


Figure 2. Regional GM volume extraction for Data_00, AAL label 1. (a) Native MRI with ROI boundary; (b) binary ROI mask; (c) its intersection with the binary GM mask; (d) full 3D voxel counting, conversion to mm³ and feature export. The slice is illustrative; volume uses all slices. Displayed feature-table values are rounded.

3.7 Feature Standardization and SVM Classification

Before classification, we standardise the 90 volume features using StandardScaler. For each feature, we subtract the mean and divide by the standard deviation calculated from the training data. This puts features with different numerical ranges on a comparable scale. We use an SVM classifier with a radial basis function (RBF) kernel, with C set to 1 and gamma set to 'scale'.

For standardized vectors z and z′, the RBF kernel is $K(z,z') = \exp(-\gamma\|z-z'\|^2)$. Thus, feature scale affects the distances used for similarity. C = 1 fixes the penalty on margin violations relative to regularization; larger C penalizes violations more strongly. Gamma controls the reach of similarity. The implementation sets gamma="scale" to $1/(p \times \mathrm{Var}(Z))$, where p = 90 and Z is the matrix supplied to SVC after scaling [8]. These fixed baseline settings were not selected on the test set. Scaling does not remove feature correlations, anatomical errors or subject-level head-size differences.

The SVM separates classes with a large margin in the kernel-defined space. Its soft margin permits violations while penalizing them; training examples with non-zero coefficients are the support vectors that determine the decision function [9]. Cortes and Vapnik [9] provide this foundational learning formulation, whereas the software documentation [8] specifies parameter behaviour. The RBF model is a nonlinear baseline; we did not establish superiority over other classifiers.

We use StratifiedKFold(n_splits=5, shuffle=True, random_state=73) on the 40 training subjects, with similar AD/NC proportions in each fold. Each round fits the complete scaler/SVM pipeline on four folds and validates on the fifth, so every subject is held out once. The scaler learns only from that round's training portion, avoiding leakage from validation values. SVM settings remain fixed; no hyperparameter search is performed. Labels are encoded NC=0 and AD=1.

After cross-validation, we fit the scaler and train the final SVM classifier using all 40 training subjects. We then apply the fitted scaler to the features of the 10 test subjects and predict their labels using the final model. The test labels are only used to evaluate the predictions and are not used for model training or parameter selection.

3.8 Descriptive Analysis of Training Features

After the original predictions, a training-only descriptive supplement examines overall GM scale and regional composition in Data_00–39. For each subject, S sums the 90 volumes and $q_r = V_r/S$. These fractions are not ICV correction. We report sample mean ± SD for S and complete binary GM volume. For all 90 labels, Cohen's $d = (\mathrm{mean}_{AD} - \mathrm{mean}_{NC})/s_p$, with $s_p^2 = (18s_{AD}^2 + 20s_{NC}^2)/38$. No ROIs are selected or significance tests performed.

Native-mask headers are audited for dimensions and spacing by training class. No test features are read or classifiers refitted. Effects describe measurements, not SVM importance. The adjustment experiments in [11] motivate examining representation, although our denominator differs. TRIPOD+AI [12] informs transparent separation of development, evaluation and later exploratory analysis, without certifying clinical validity.

4. Results

4.1 Brain Extraction Results

Figure 3 compares brain extraction for Data_00 using the default and selected BET parameters. With the default parameters, neck and facial tissue remains in the sagittal and coronal slices. The selected settings remove much of this non-brain tissue, particularly in Sagittal #84 and Coronal #104. The difference is smaller in the higher axial slice, Axial #167. The displayed slices therefore support improved removal of non-brain tissue at these positions, but do not establish extraction accuracy throughout the image. Figure B1 provides enlarged views of the selected result at the same three slice positions.

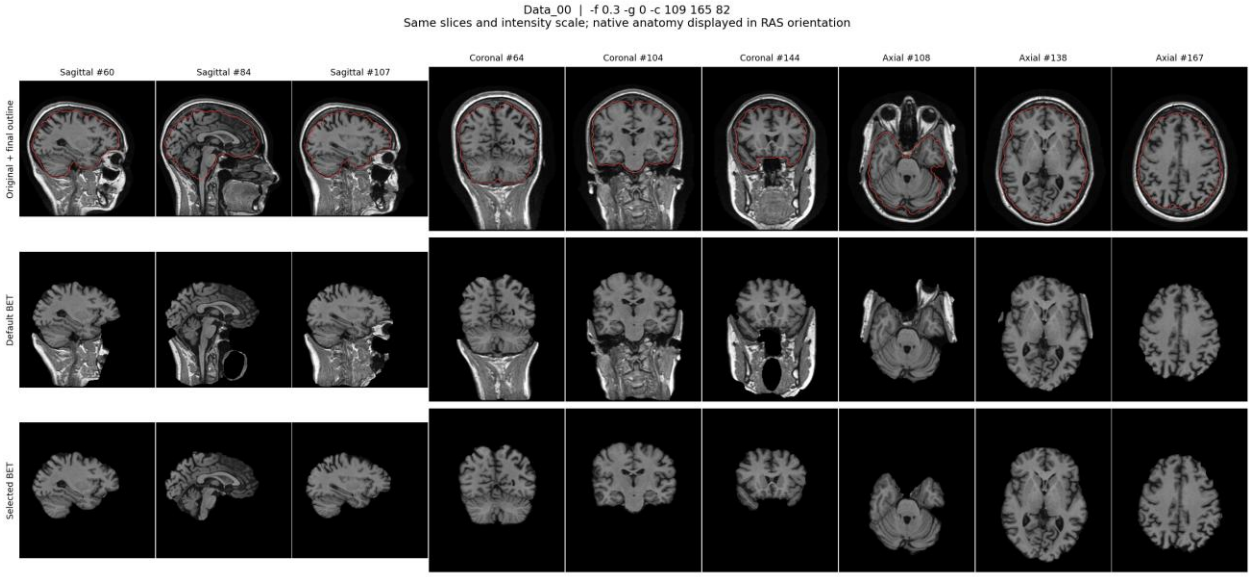


Figure 3. BET brain extraction for Data_00. Rows show the original image with the selected mask boundary, default BET extraction and selected-parameter BET extraction.

4.2 Tissue Segmentation and Grey Matter Images

Figure 4 shows tissue segmentation for Data_00. The red, cyan and yellow outlines mark the boundaries of the grey matter, white matter and cerebrospinal fluid masks, respectively. In these slices, the outlines generally follow the visible tissue structures, allowing inspection of the spatial distribution of the three classes. This visual agreement does not establish accurate segmentation at every boundary or throughout the volume. Figure B2 displays the GM intensity image separately: the tissue-mask outlines show binary class boundaries, whereas the masked intensity image retains the MRI intensities within the GM mask.

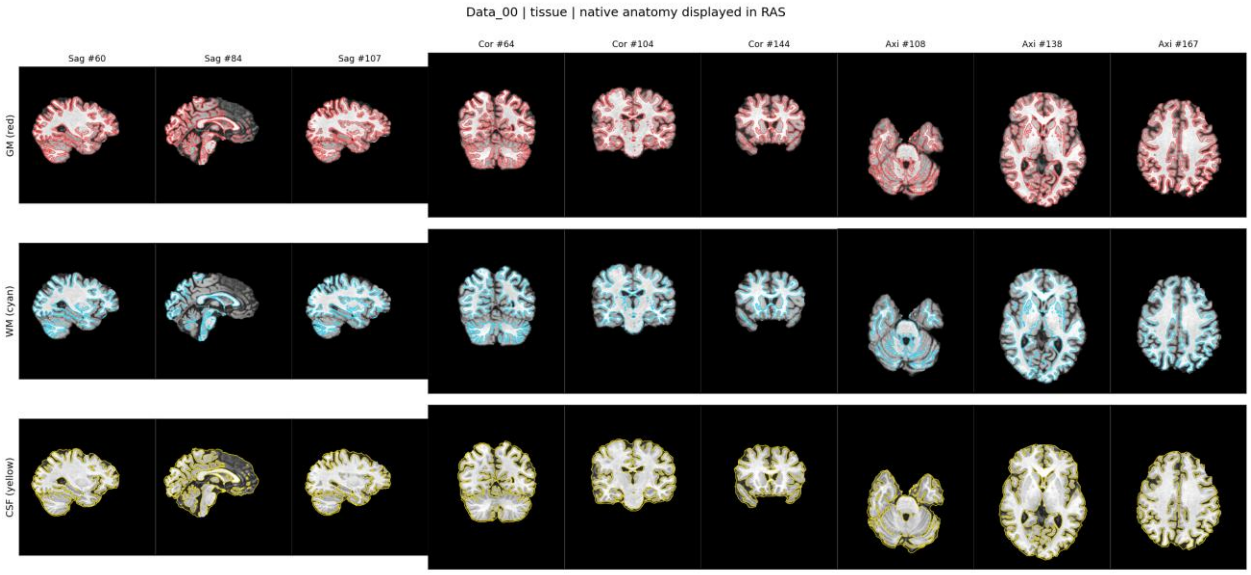


Figure 4. FAST tissue segmentation for Data_00. Rows show grey matter (red), white matter (cyan) and cerebrospinal fluid (yellow) binary-mask boundaries over the anatomical image.

4.3 Registration and Atlas Alignment Results

Figure 5 shows the affine registration, nonlinear registration and transformed AAL atlas for Data_00. In the first row, local misalignment is visible in Sag #84 and Sag #107, where the template does not fully match the underlying brain outline. Similar differences are visible in Cor #64 and Cor #104, while Cor #144 and the axial slices appear better aligned. In the second row, the nonlinear registration appears to improve the alignment in these areas, and the template follows the brain outline more closely. In the third row, the AAL labels generally lie within the brain outline. However, a small area at the bottom of Cor #64 remains visible without a colored label. This alone does not confirm a registration error, as some areas may not be labelled by the atlas. These observations are based on the displayed slices rather than a quantitative assessment of registration accuracy. Larger views of Sagittal #84 and Coronal #104 are provided in Figure B3.

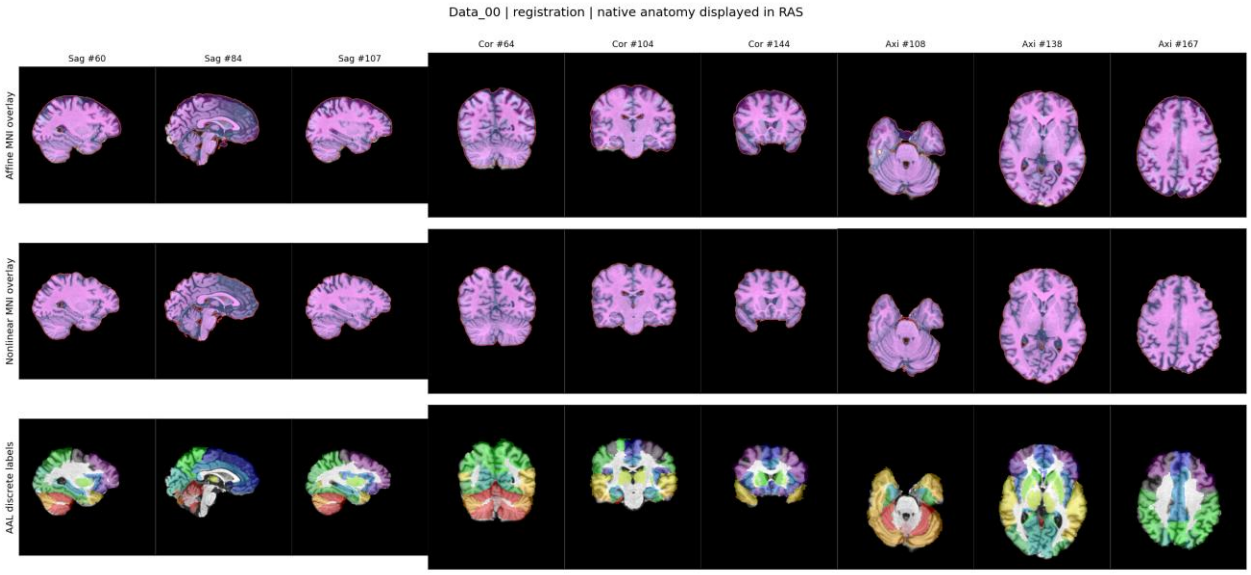


Figure 5. Registration and atlas alignment for Data_00. Rows show the affine template, nonlinear template and transformed AAL labels in native space, across sampled sagittal, coronal and axial slices.

4.4 Regional Grey Matter Volume Features

After regional grey matter volume extraction, we obtain 90 volume features for each subject, forming one feature vector. All volumes are measured in mm^3. The training and test feature matrices have dimensions of 40×90 and 10×90, respectively, giving a total of 4,500 volume measurements. Table 2 shows the grey matter volumes of the first five AAL regions for three training subjects. The complete results are saved in the spreadsheet submitted with the report (Project01_90_ROI_GM_volumes.xlsx). These features are then used for standardization and SVM classification. Subject IDs are not included as input features.

Table 2. Example regional grey matter volumes for selected training subjects (mm^3).

Subject ID	ROI 1	ROI 2	ROI 3	ROI 4	ROI 5
Data_00	6523.22	4795.51	6533.90	6715.43	1947.68
Data_01	7723.00	7983.00	9151.00	9135.00	2659.00
Data_02	8852.00	8397.00	5811.00	9491.00	2347.00

Values are rounded to two decimal places for display. The complete measurements for all 50 subjects and 90 regions are provided in the accompanying spreadsheet.

All 4500 exported volumes were checked independently by counting non-zero GM voxels within each ROI and multiplying by voxel volume. The stored checks agree with the FSL measurements; the maximum absolute difference is approximately 0.000906 mm^3, consistent with numerical precision differences. No ROI among labels 1–90 has a zero volume. File and geometry checks cover all 50 subjects. Manual review examined sampled BET views for all subjects, while sampled FAST/registration review covered 10 subjects. These checks support numerical consistency and identify visible limitations; they do not establish anatomical accuracy throughout every scan. Appendix A.5 identifies the verification records and sampled subjects.

4.5 Cross-Validation Performance

Figure 6 shows the results of five-fold cross-validation on the 40 training subjects. The accuracies of the five folds are 100%, 100%, 100%, 87.5% and 100%, respectively. The mean accuracy is 97.5%. Four folds correctly classify all validation subjects, while one fold has a lower accuracy. These results show good classification performance during cross-validation. However, they are based on the training set, and the final test results are reported in Section 4.6. The standard deviation across the five fold accuracies is 5.0 percentage points; it describes variation across these folds, not a confidence interval.

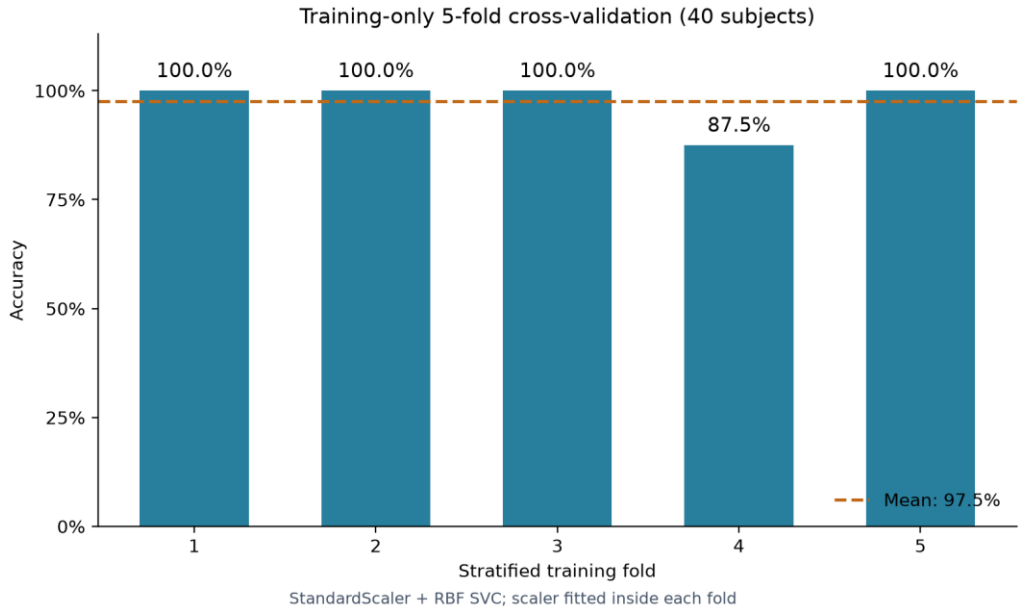


Figure 6. Five-fold cross-validation accuracy within the training set. The fold accuracies are 100%, 100%, 100%, 87.5% and 100%, with a mean of 97.5%.

These scores evaluate validation folds within the 40-subject training set. They are separate from the final held-out test result.

4.6 Test Set Predictions and Performance

Table 3 shows the predicted and true labels for the 10 test subjects, and Figure 7 shows the confusion matrix. The model correctly classifies 9 out of 10 subjects, giving a test accuracy of 90%. All four NC subjects are correctly classified, while five out of six AD subjects are correctly classified. This gives an AD sensitivity of 83.3% and an NC specificity of 100%. The only misclassified subject is Data_49, whose true label is AD but is predicted as NC. The test accuracy is lower than the mean cross-validation accuracy of 97.5%.

Table 3. Predicted and true labels for the test subjects.

File name	Predicted label	True label
Data_40.nii	NC	NC
Data_41.nii	NC	NC
Data_42.nii	AD	AD
Data_43.nii	NC	NC
Data_44.nii	AD	AD
Data_45.nii	NC	NC
Data_46.nii	AD	AD
Data_47.nii	AD	AD
Data_48.nii	AD	AD
Data_49.nii	NC	AD

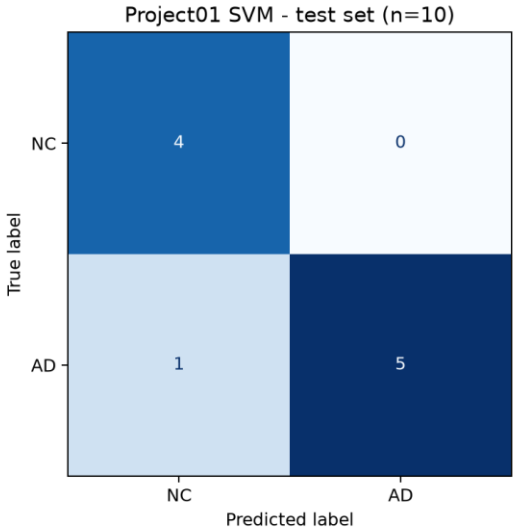


Figure 7. Held-out test confusion matrix. Rows indicate true labels and columns indicate predicted labels, ordered NC and AD. Four NC and five AD subjects were correctly classified; one AD subject was predicted as NC.

4.7 Training Feature Scale and Image Geometry

Table 4 summarizes the training volumes. The mean AAL 1–90 sum is 20.9% lower in AD than NC, with overlapping individual ranges (Figure 8a). Absolute-volume d is negative for 79/90 ROIs. After expressing volumes as fractions of their subject's AAL sum, 48/90 effects are negative and 31 change sign. Median |d| changes from 1.22 to 0.66 (Figure 8b). These are descriptive contrasts, not 90 independent findings of disease effects.

Table 4. Training GM volumes in cm^3 (sample mean ± SD; 1 cm^3 = 1000 mm^3).

Measure	NC (n = 21)	AD (n = 19)
Complete binary GM mask	604.8 ± 58.2	474.4 ± 58.6
AAL labels 1–90 sum	437.8 ± 44.6	346.4 ± 46.9

All 21 training NC images have a 256 × 256 × 176 grid and 1 × 1 × 1 mm voxels. All 19 AD images have 256 × 256 × 166 or 170 grids and non-isotropic voxels (approximately 0.94 × 0.94 × 1.20 mm). Thus, stored geometry is class-associated. Geometry alone does not identify scanner or site provenance. The training summaries and header audit are retained with the analysis code in the repository.

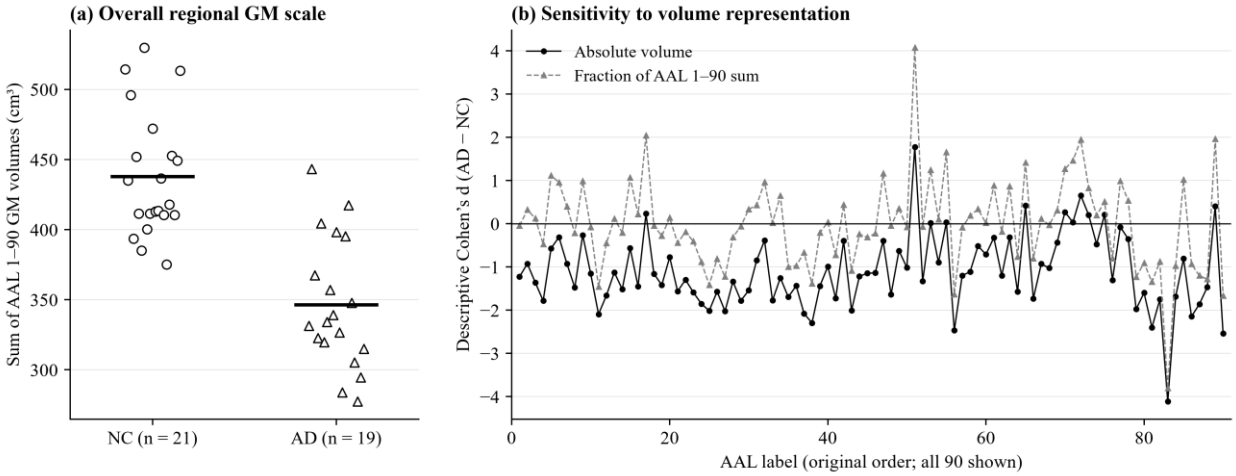


Figure 8. Post hoc description of training features only. (a) Each point is one subject's AAL 1–90 GM sum; horizontal bars indicate class means. (b) AD-minus-NC standardized mean differences for all 90 labels, using absolute volumes or fractions of each subject's AAL sum. Connecting lines aid tracing label order. Fractions are not ICV correction, and effects are not SVM importance estimates.

5. Discussion and Conclusion

5.1 Interpretation of the Findings

The pipeline correctly classifies nine of ten held-out subjects, supporting discrimination within the supplied dataset. This predictive result does not establish that the classifier detects disease-specific regional atrophy. Section 4.7 shows that lower overall regional GM volume and a class-associated difference in image geometry accompany the labels. These observations make it necessary to consider several explanations for the measured separation before interpreting it biologically.

The validation percentages need their denominators. Each CV fold contains eight subjects: one fold has one error, giving 39/40 correct across held-out folds. The test also has one error, among ten subjects. The 7.5-percentage-point gap therefore accompanies coarse accuracy increments and different samples; it neither establishes overfitting nor equivalence of performance. The 5-percentage-point fold standard deviation describes this partition, not uncertainty in population performance.

Karas et al. [1] support the biological rationale through regional GM loss; Klöppel et al. [2] demonstrate MRI/SVM classification with a voxel-based representation. Our regional summaries retain atlas identities but can hide local changes represented by voxel features. The 2024 review [4] covers different disease stages and populations, and its pooled prevalence estimates are not an accuracy benchmark. These sources motivate the experiment without establishing comparability of performance or identifying which ROIs drive our predictions.

The training analysis makes the distinction between feature scaling and anatomical adjustment concrete. AAL-summed GM is 20.9% lower on average in AD, and 79 regional absolute-volume effects are negative; much of the pattern therefore accompanies a broad volume-scale difference. Expressing measurements as fractions reduces median |d| and reverses 31 directions. A larger fraction can arise when the denominator decreases more than the numerator, so a positive relative effect does not demonstrate tissue growth. Because the fractions sum to one, they also constrain one another. Neither representation establishes the contribution of an ROI to the nonlinear SVM.

Barnes et al. [10] show why head size matters for volumetric interpretation. Voevodskaya et al. [11] go further by comparing raw, proportional and residual ICV adjustment in PIVUS and ADNI: adjustment changes regional relationships and can affect diagnostic comparisons. Their FreeSurfer measurements and ICV denominator differ from our binary AAL volumes and GM-sum denominator, so their results support examining the adjustment choice, not adopting our fractions as a validated correction. Global GM loss may itself contain disease information; removing scale could discard it. Without ICV, demographics and acquisition provenance, the current analysis cannot partition disease effects, baseline anatomy and measurement differences.

With 90 features and 40 training subjects, alternative feature selection, head-size adjustment or SVM tuning would require a separate training-only assessment, such as nested validation. No such comparisons were performed here. Any later changes made after inspecting these test results

should be reported as exploratory and evaluated on a new independent test set, rather than repeatedly selecting the best result on these ten subjects.

The geometry audit shows that stored image geometry differs systematically between the diagnostic groups. Stratifying subjects preserves diagnostic balance, but both classes retain their characteristic stored geometry across folds; the validation design therefore cannot isolate disease from geometry-associated acquisition or processing effects. Converting counts to mm^3 accounts for voxel volume, yet does not equalize partial-volume behaviour or segmentation boundaries. Geometry was not supplied as an SVM feature, and this audit does not prove that it caused the predictions. A dataset containing AD and NC scans with overlapping acquisition conditions is needed to distinguish these possibilities; merely increasing the number of random folds would not resolve them.

5.2 Limitations and Potential Improvements

Figure 9 shows remaining boundary problems in the adopted BET result for Data_44: suspected upper-frontal brain omission and residual non-brain tissue. These observations matter because an omitted brain voxel is unavailable to subsequent GM segmentation, while retained non-brain material can complicate tissue classification. Registration differences can also move the ROI boundary used for counting. These are plausible routes by which preprocessing affects features, not measured explanations of an individual prediction. Data_44 is correctly classified as AD, which does not validate its anatomical processing. Figure B4 enlarges the retained boundary; further boundary review would be needed before revising this subject's measurements.

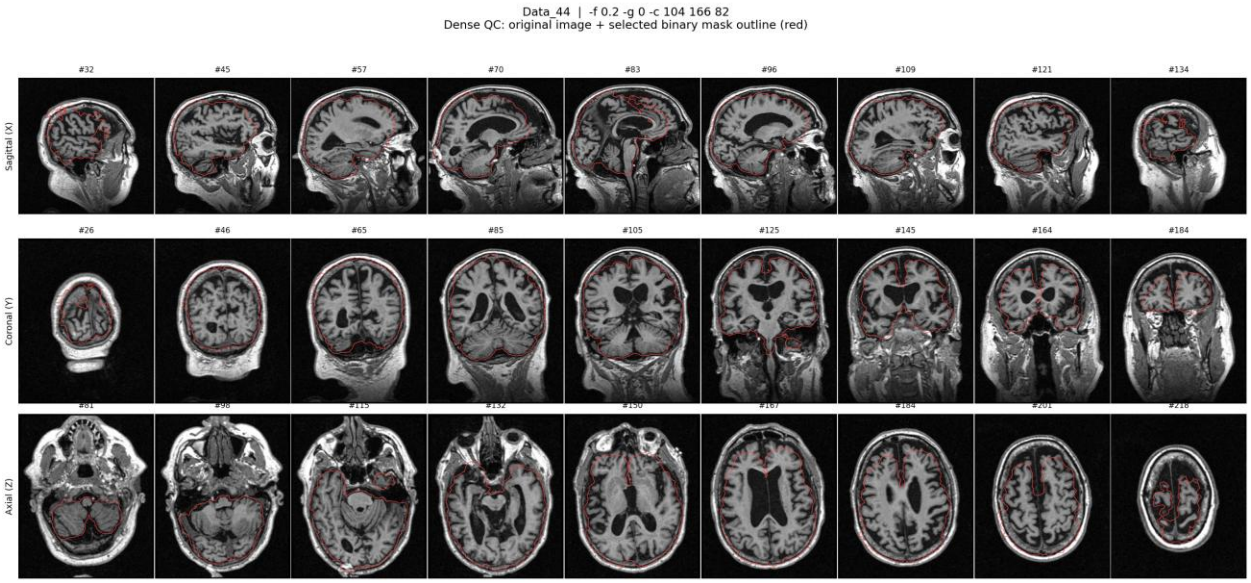


Figure 9. Multi-slice review of the retained BET result for Data_44. The repository quality review identifies suspected frontal boundary omission and residual non-brain tissue; this case illustrates a preprocessing limitation.

The binary GM masks also impose a hard tissue decision at each voxel. A boundary voxel changing between zero and one changes the count, so segmentation and ROI-alignment uncertainty

can affect regional volumes even when the arithmetic is reproducible. Probability-weighted GM measurement was not evaluated; it would represent a different feature definition and should not be treated as an improvement established by this experiment.

Finally, NC specificity of 100% is based on only four correct NC predictions, and AD sensitivity of 83.3% on five of six AD subjects. One additional test error changes overall accuracy by ten percentage points. These small denominators limit claims about performance in other populations. Data_49 remains the sole false-negative AD case; its cause cannot be established from the current results or attributed to the separate Data_44 observations. The fixed course test set is held out from fitting, but it is not an external clinical validation cohort.

The independent count comparison verifies that the specified masks produce the exported volume values. Sampled image review assesses visible anatomical plausibility, including the unresolved Data_44 boundary issue, while held-out predictions measure agreement with labels in this dataset. These checks support different conclusions. Data_44 can be classified correctly despite an imperfect mask, and a reproducible volume can still include a misplaced boundary. The published evaluations of BET, FAST and registration [5–7] explain the methods' basis, but do not establish their accuracy on the images processed here.

The practical priority is to audit acquisition provenance and obtain diagnosis groups with overlapping geometry, alongside age, sex and ICV, before testing a prespecified adjustment strategy. Image boundaries and registration would then be reviewed with anatomical references, with remaining changes assessed inside training-only validation and a new test cohort. Following the performance distinctions in TRIPOD+AI [12], our fixed-label accuracy, sensitivity and specificity do not establish calibrated disease probabilities or clinical utility. SVM decision scores are not probabilities; no calibration or decision-benefit analysis was performed. The present work demonstrates a course classification pipeline rather than a clinical diagnostic instrument.

5.3 Conclusion

This project implements MRI preprocessing, 90 native-space regional GM volumes and an SVM in a reproducible workflow. Numerical checks confirm consistency of the exported features, while sampled images document remaining anatomical limitations. Training-only five-fold validation gives 97.5% mean accuracy, and the fixed test set gives 9/10 correct, with 83.3% AD sensitivity and 100% NC specificity. The added training analysis shows broad GM-scale differences and class-associated image geometry. Accordingly, the results demonstrate dataset-specific discrimination, while acquisition-controlled validation and anatomical/covariate assessment are needed to establish how much separation reflects AD-related tissue changes.

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Appendix A Representative commands and settings

The commands below mirror the completed batch scripts for Data_00. Run them from the project root in the configured Linux VM with FSL and NiftyReg on PATH. They illustrate the existing processing; the four main scripts apply the same sequence to all subjects using their own BET parameters.

A.1 Tissue segmentation and grey matter masking

```
b=Output/train/Data_00
tpl=Packages/MNI152_T1_1mm_brain.nii.gz
aal=Packages/aal.nii.gz

fast -S 1 -n 3 -t 1 -g -v -o "${b}_brain" "${b}_brain.nii.gz"
mv "${b}_brain_seg_0.nii.gz" "${b}_csf_mask.nii.gz"
mv "${b}_brain_seg_1.nii.gz" "${b}_greymatter_mask.nii.gz"
mv "${b}_brain_seg_2.nii.gz" "${b}_whitematter_mask.nii.gz"
fslmaths "${b}_brain.nii.gz" -mul "${b}_greymatter_mask.nii.gz" \
  "${b}_brain_greymatter.nii.gz"
```

The options specify one input channel (-S 1), three tissue classes (-n 3), a T1-weighted image (-t 1), binary masks (-g), diagnostic output (-v), and an output base name (-o). Class indices 0, 1 and 2 are renamed CSF, GM and WM. Multiplication retains brain intensities where the GM mask is one and sets all other voxels to zero.

A.2 Template registration and AAL label resampling

```
reg_aladin -ref "${b}_brain.nii.gz" -flo "$tpl" \
  -aff "${b}_MNI_to_native_affine.txt" -res "${b}_MNI_affine.nii.gz"
reg_f3d -ref "${b}_brain.nii.gz" -flo "$tpl" \
  -aff "${b}_MNI_to_native_affine.txt" \
  -cpp "${b}_MNI_to_native_cpp.nii.gz" -res "${b}_MNI_warped.nii.gz"
reg_resample -ref "${b}_brain.nii.gz" -flo "$aal" \
  -trans "${b}_MNI_to_native_cpp.nii.gz" -inter 0 \
  -res Output/train/AAL_to_Data_00.nii.gz
```

The affine matrix initializes nonlinear refinement. The original template is used again, avoiding a second application of the affine transform to an already warped template. The AAL atlas is resampled with nearest-neighbour interpolation (-inter 0), which preserves integer region labels rather than averaging their IDs.

A.3 Measuring grey matter volume in one ROI

This example isolates native AAL label 1. In the batch workflow, the unchanged CreateSeedMask helper creates the masks and Measurement iterates over labels 1–90.

```
fslmaths Output/train/AAL_to_Data_00.nii.gz -thr 1 -uthr 1 -bin \
  Output/train/Data_00_ROI_1_example.nii.gz
fslstats Output/train/Data_00_greymatter_mask.nii.gz \
  -k Output/train/Data_00_ROI_1_example.nii.gz -V
```

Thresholding retains label 1 and -bin produces its binary ROI. The -k option restricts binary GM counting to that ROI. The -V output gives voxel count then mm³, measuring regional GM rather than the complete anatomical ROI. This case gives 6523.223317 mm³, as in Table 2.

A.4 Reproducing the SVM validation settings

These settings use the loaded 90-feature matrices X_train and X_test and labels y_train, aligned by subject ID. IDs are excluded as features. The notebook and project01_svm.py contain the loading and verification steps.

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.model_selection import StratifiedKFold, cross_val_score

model = make_pipeline(StandardScaler(),
                      SVC(kernel="rbf", C=1, gamma="scale"))
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=73)
scores = cross_val_score(model, X_train, y_train,
                         cv=cv, scoring="accuracy")
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Each CV fold fits its own scaler/SVM, excluding validation values from fitting. The final pipeline uses all 40 training subjects. Predictions are saved before test-label comparison. The implementation and recorded original run are retained at https://github.com/Babe153/Project01_5622.

A.5 Processing verification records

Checks cover all 50 subjects and labels 1–90. Sampled FAST/registration review covers Data_00, 01, 15, 18, 33, 35, 36, 40, 44 and 49; it is not full-volume anatomical ground truth.

```
report_evidence/validation/volume_verification.json
report_evidence/validation/visual_review.json
submission/processing_qc_records.csv
```

A.6 Reproducing the training supplement

From the repository root, run this utility with numpy, matplotlib and nibabel. Training-only outputs accompany Figure 8.

```
python report_evidence/figure_code/analyse_training_volumes.py
```

Appendix B Enlarged views of existing processing results

These views use the same final NIfTI outputs as the main figures. They enlarge selected slices without changing the processing. Slice indices refer to RAS-oriented display arrays.

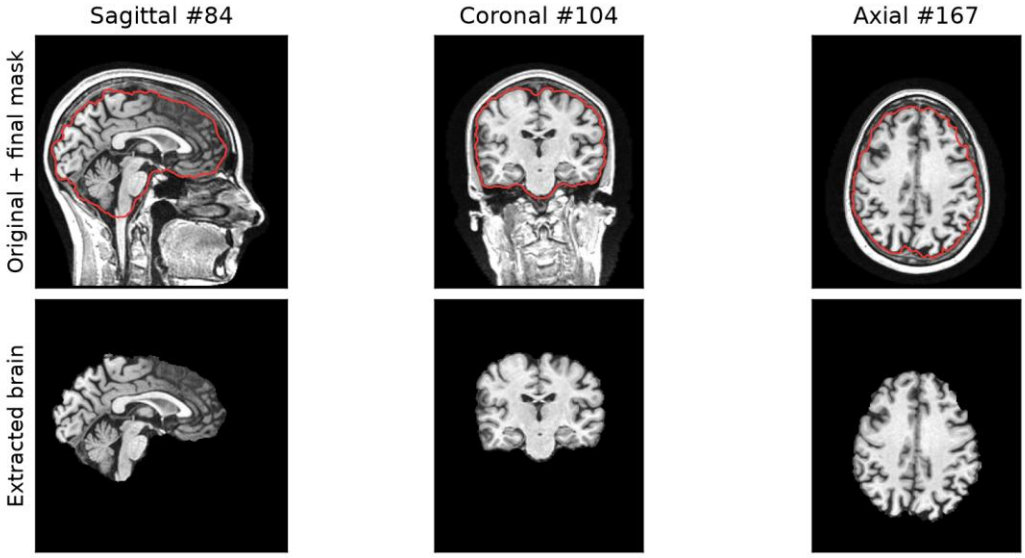


Figure B1. Selected BET result for Data_00 at the slice positions discussed in Section 4.1. Top: original MRI with final mask boundary in red. Bottom: extracted brain. These are the final settings, not the default BET result.

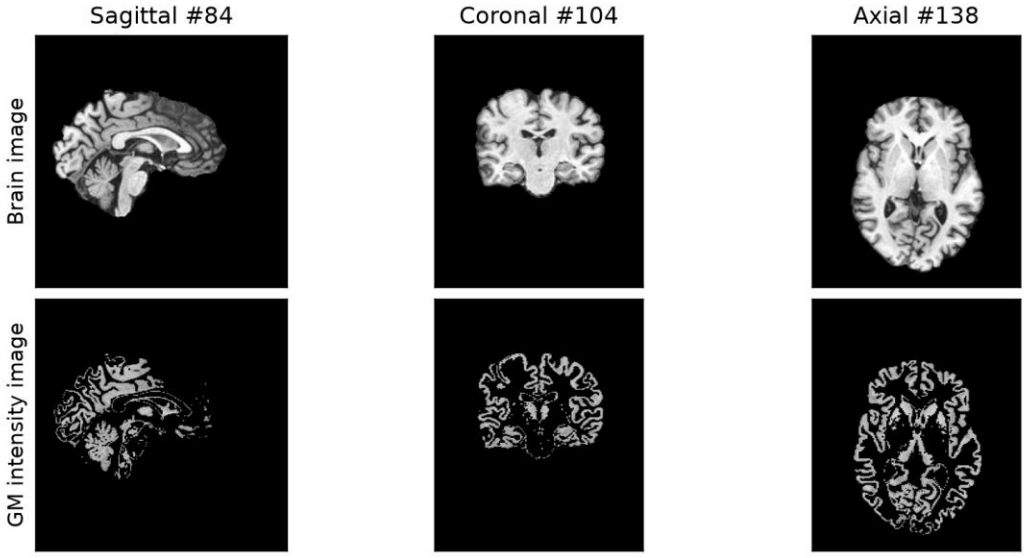


Figure B2. Data_00 brain image (top) and the GM intensity image obtained by multiplying it by the binary GM mask (bottom). The same intensity scale is used in both rows. The GM mask, rather than image brightness, defines the measured volume.

B.1 Registration and the retained BET limitation

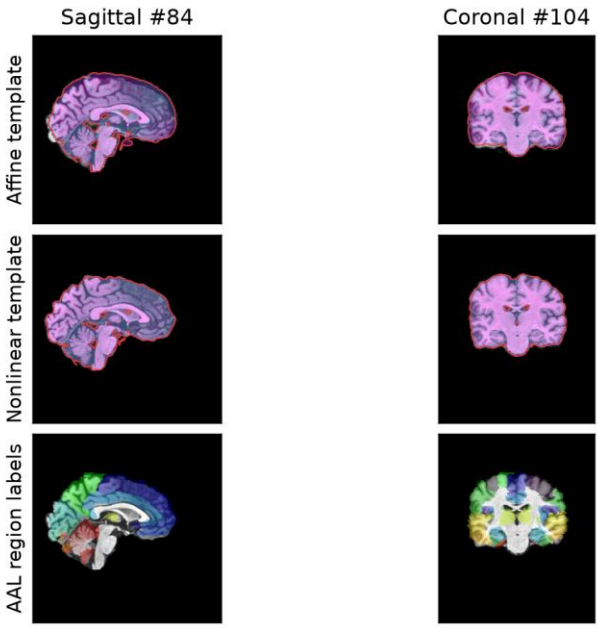


Figure B3. Larger views of Data_00 at Sagittal #84 and Coronal #104. Rows show affine template overlay, nonlinear template overlay and resampled AAL labels on the subject brain. This is a visual comparison, not a quantitative registration score.

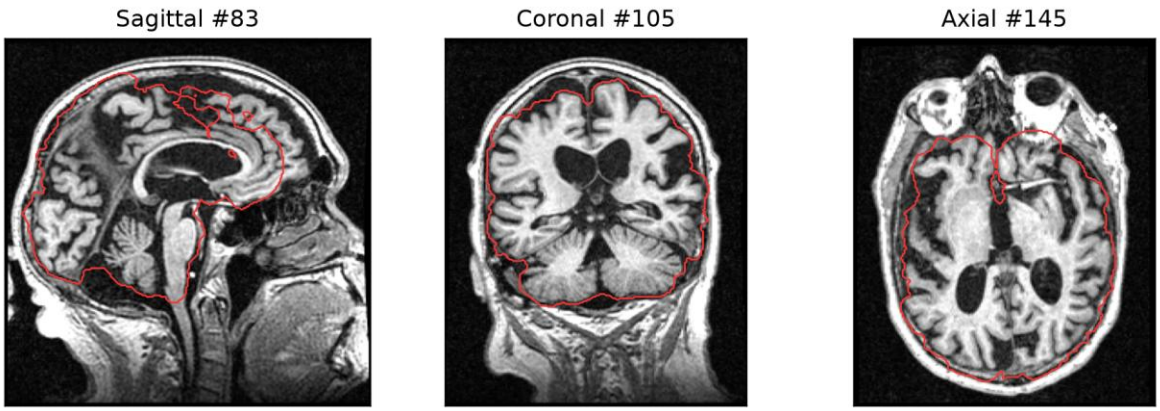


Figure B4. Retained Data_44 BET mask boundary (red) over the original MRI at three sampled positions. Suspected upper frontal tissue omission is visible in the sagittal view. These views illustrate the limitation discussed in Section 5.2; they do not establish its effect on classification.